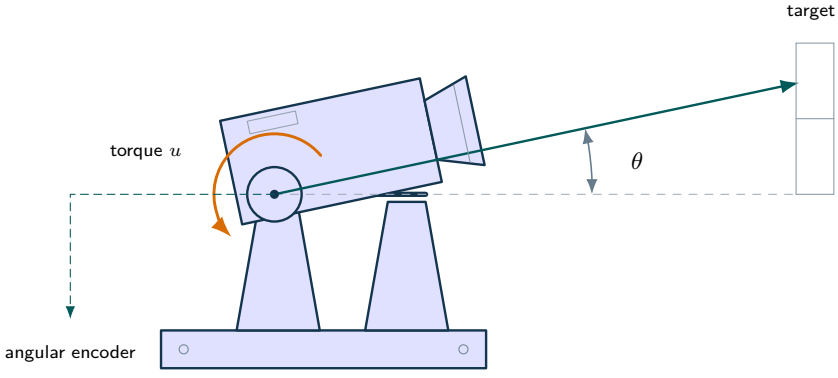

Gimbal axis with angular readout

Normalized model $G(s) = 1/[s(s+1)]$; geometry is illustrative

Coordinate and actuation

Positive angle is measured from the dashed reference ray. Torque acts about the marked gimbal pivot.

Normalized axis

$G(s) = \dfrac{1}{s(s+1)}$.

Lead supplies phase; lag changes low-frequency gain and the ramp error.

Limits of the view

Encoder and lens establish the signal meaning. Camera inertia, gearing and dimensions do not identify this stipulated plant.